FastSpeech 2 Text-to-Speech for English and Telugu

Technical Report

Code, results and checkpoints: https://github.com/hpi52003/Fastspeech2-English-Telugu

1. Implementation Approach and Modifications

FastSpeech 2 was trained separately for English and Telugu on about 5 hours of training audio per language, and the predicted duration, pitch and energy were analysed on a fixed set of test sentences. The implementation starts from the reference repository linked in the assignment (github.com/ming024/FastSpeech2). The model is unchanged: a transformer encoder, a variance adaptor with duration, pitch and energy predictors and a length regulator, and a mel decoder with a postnet. It is trained with five loss terms: mel before the postnet, mel after the postnet, pitch, energy and duration. The modifications concern the data pipeline, which differs between the two languages.

FastSpeech 2 requires a phone-level alignment for every training utterance, so the text-to-phone-to-alignment pipeline was the main part of the work. English reuses the ARPAbet phone set and a pretrained Montreal Forced Aligner (MFA) model. Telugu had no such front end, so a grapheme-to-phoneme mapper, a pronunciation dictionary and an aligner trained on the Telugu corpus itself were built.

Changes compared with the reference code

Part	English	Telugu
Text front end	ARPAbet phones from the MFA english_us_arpa dictionary. Numbers are spelled out before alignment. Out-of-dictionary words use MFA's G2P model for training and g2p_en at inference.	New rule-based grapheme-to-phoneme mapper (telugu_prep.py) with a 47-phone inventory: short and long vowels, aspirated and retroflex stops, geminates, and a nasal that assimilates to the following consonant. Numerals are expanded into Telugu words at inference.
Alignment	Pretrained MFA acoustic model english_us_arpa.	MFA acoustic model trained from scratch on the 2,762 selected utterances (monophone, triphone, LDA, three SAT passes; about 2 h 18 min).
Speakers	Multi-speaker (115 speaker ids, see section 2).	Single speaker.
Training loop	Added --resume, validation every 1,000 steps on the full validation set, loss logging to CSV every 100 steps, best-checkpoint selection by validation loss, and upload of checkpoints and logs to the Hugging Face Hub so that an interrupted Kaggle session does not lose progress.	
Config	Separate config.yaml per language (paths, phone set, speaker count, statistics files).	

2. Data

2.1 Selection and subsampling

Both datasets were inspected (hours, clip-length percentiles, sampling rate, speakers) before subsampling. Audio was resampled to 22.05 kHz mono and peak-normalised. For both languages the random seed was 42; the pool was filtered, then about 1 hour was drawn for validation and about 5 hours for training.

	English (Svarah)	Telugu (IndicTTS Telugu)
Source	ai4bharat/Svarah, the only split on the Hub (test), used as the pool	SPRINGLab/IndicTTS_Telugu, train split

	English (Svarah)	Telugu (IndicTTS Telugu)
Voice(s)	Many speakers of accented English	One voice only (gender label 1), selected after listening to clips from both labels
Filters	1.5–15 s; letters, digits and basic punctuation only; at least 3 words; 5–25 characters per second	2–15 s; Telugu Unicode block plus basic punctuation; 5–30 characters per second; duplicate sentences removed
Split unit	Recording group (first token of the file name), so no group appears in both sets	Individual utterance; no sentence appears in both sets
Extra checks	Every validation speaker also exists in training; no validation sentence (case-insensitive) occurs in training	Assertion that utterance ids and texts do not overlap
Used after preprocessing	3,133 train / 589 val utterances 4.65 h train / 0.92 h val	2,306 train / 456 val utterances 4.27 h train / 0.86 h val

Both training sets are below 5 hours after preprocessing because the silence at the start and end of each clip is trimmed using the alignments (raw selection: 5.01 h train / 1.00 h val for English, 5.0 h / 1.0 h for Telugu). For English, 16 further utterances were dropped because the aligner produced a phone longer than 1.5 s, which usually indicates a bad alignment.

Speakers in Svarah. Svarah has no speaker-id column and the file names do not provide one. A speaker was therefore defined as a unique combination of demographic fields (gender, age group, mother tongue, native state and district, qualification, job category, occupation domain), which gives 115 ids in the training set. This is an approximation, since different people with the same profile share one id.

2.2 Preprocessing

English. The first alignment run failed on about 40 transcripts containing numbers (years such as 1990s, ordinals, decimals, numbers with commas). A normaliser for these cases was written, with a fallback that reads any remaining digit as a word, after which all utterances aligned.

Telugu. The mapper produced 13,195 distinct words and used 46 of the 47 phones (rr does not occur; the rarest are jh with 4 occurrences, ddh with 26 and chh with 45). The TextGrids were checked before training: no spn labels, median longest phone 0.21 s, maximum 1.38 s, none above 1.5 s.

Features. Feature extraction follows the reference code: mel spectrograms at 22.05 kHz, phone durations from the TextGrids, F0 with PyWorld and energy from the STFT. Pitch and energy statistics (stats.json) are computed on the training set and used for normalisation.

2.3 Training configuration

	English	Telugu
Training steps	30,000	15,000
Batch size	about 64 (inferred from 49 and 37 iterations per epoch)	
Validation	every 1,000 steps on the full validation set; best checkpoint kept by total validation loss	
Wall-clock time	5 h 47 min (about 1.44 it/s)	3 h 02 min (about 1.38 it/s)
Best checkpoint used	step 12,000 (val loss 2.608)	step 15,000 (val loss 2.066)
Optimiser, learning rate, warm-up, dropout	Adam with learning-rate warm-up and step-wise annealing, as in the reference code; exact values are in fastspeech2/config.yaml	
Training hardware	Kaggle notebook GPU (Tesla T4); inference on a Colab T4	

	English	Telugu
Vocoder for the test audio	Pretrained HiFi-GAN vocoder from the reference code, not trained or fine-tuned	

3. Training and Validation Loss

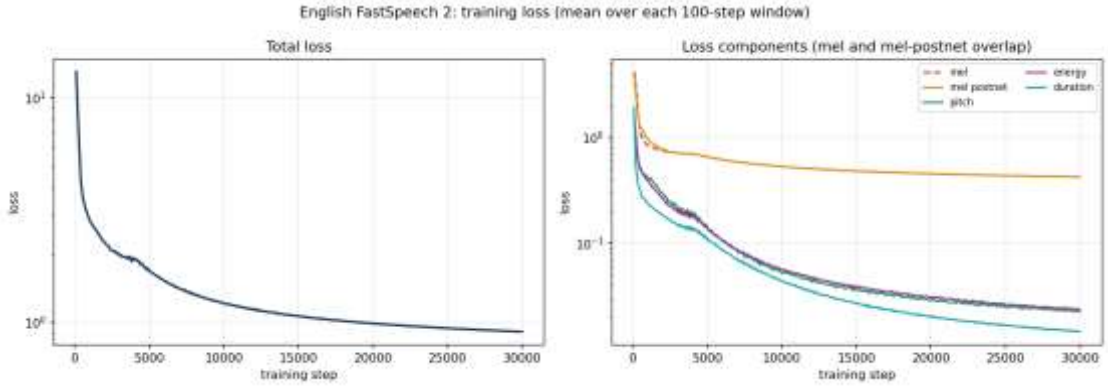


Figure 1. English training loss.

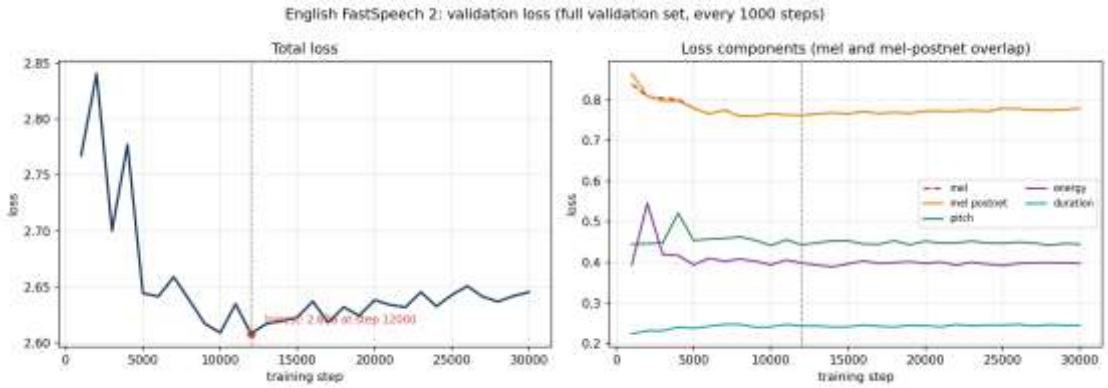


Figure 2. English validation loss. The dashed line marks the best checkpoint (step 12,000).

English

The training loss falls from 13.06 at step 100 to 1.37 at step 7,600, 1.06 at step 15,100 and 0.91 at step 30,000, and keeps decreasing. Validation loss starts at 2.77 and rises briefly to 2.84 at step 2,000 and 2.78 at step 4,000 (driven by the energy and pitch terms respectively). From step 5,000 it stays between 2.61 and 2.66, with its minimum of 2.608 at step 12,000, and then increases slightly to 2.645 at step 30,000 (+1.4%). Over the same period the validation mel loss rises from 0.762 to 0.779 while the training mel loss keeps falling (0.42 at the end).

This indicates mild overfitting after about 12,000 steps, which is expected for 4.65 hours of accented, multi-speaker speech; the step-12,000 checkpoint was therefore used for synthesis. The gap is largest for the variance predictors: at step 30,000 the training pitch loss is 0.023 against 0.445 on validation, and the validation duration loss stays flat at about 0.24. These predictors fit the training utterances but generalise poorly.

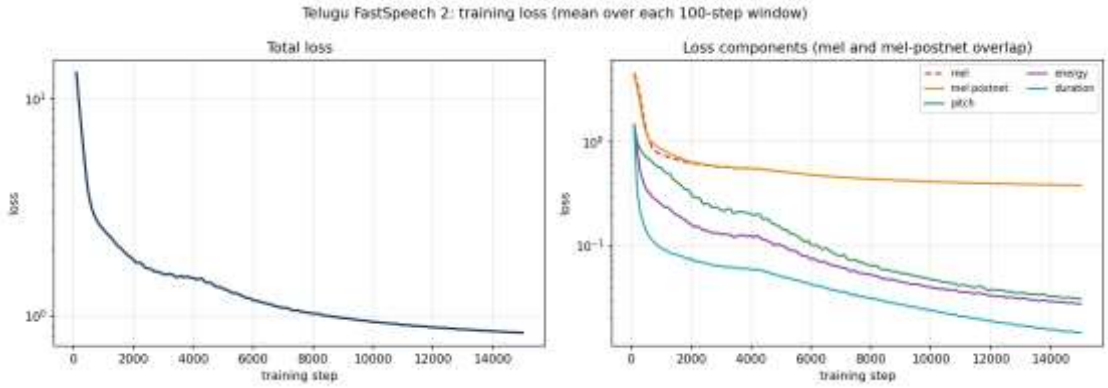


Figure 3. Telugu training loss.

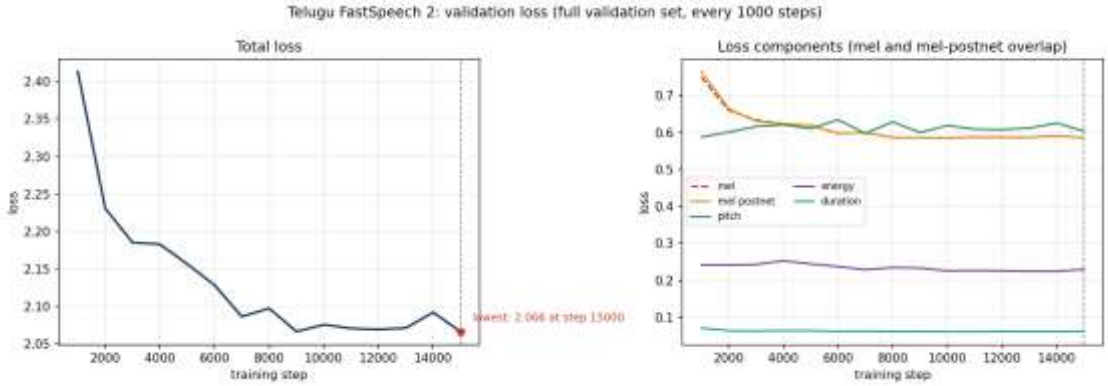


Figure 4. Telugu validation loss.

Telugu

Validation loss falls from 2.41 at step 1,000 to 2.19 at step 3,000 and 2.07 at step 9,000, then stays between 2.066 and 2.092 up to step 15,000, where the best checkpoint lies. The validation mel loss is flat at 0.586 from step 8,000. Training ended at 15,000 steps because the Kaggle GPU quota was exhausted; the plateau over the last 6,000 steps suggests limited further gain, but longer training was not tested. The validation duration loss is very low (0.061, against 0.24 for English), consistent with a single speaker and an aligner trained on the same audio.

The validation pitch loss did not improve: it is 0.587 at step 1,000 and 0.604 at the end (range 0.60–0.63), while the training pitch loss falls to 0.031. The pitch predictor fits the training utterances without generalising, which is consistent with the wide F0 contours in section 5.

4. Generated Speech: Pronunciation and Quality

Audio was generated with the best checkpoint of each language (English step 12,000, Telugu step 15,000) and English speaker id 0. The test sentences are E1, E2a, E2b, E3, E4 and T1, T2a, T2b, T3, T4. The observations below are based on the phone sequences given to the model, so front-end errors are visible directly, and on the variance-adaptor plots in section 5. No formal listening test was run (see section 6).

English

E1 – The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

The long word is converted into 28 phones with a plausible stress pattern: secondary stress (AE2) on the first and middle syllables and primary stress (EH1) on the syllable before the ending. The remaining words are ordinary dictionary words and show no front-end problems.

E2a / E2b – a) The orchestra's conduct during the ceremony was praised by the audience. b) The researchers will conduct an experiment to evaluate the proposed method.

This is the homograph test. In (a) conduct is a noun with stress on the first syllable (K AA1 N D AH0 K T); in (b) it is a verb with stress on the second syllable (K AA0 N D AH1 K T). The front end resolves both correctly, probably through the part-of-speech based homograph handling of g2p_en (not tested separately). The pronunciation is fixed before FastSpeech 2 receives the text, so the model itself plays no part in this.

E3 – The rhythm of the orchestra changed unexpectedly.

The text is used exactly as given in the assignment, including the stray space in "rhyth m". This causes a pronunciation failure: "rhyth" becomes R AY1 DH and the isolated "m" is read as a letter name (EH1 M), giving approximately "rye-th em". With the space removed (E3_fixed), "rhythm" becomes R IH1 DH AH0 M, which is correct. The failure is therefore caused by the input, not by the model.

E4 – The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

The normaliser expands the sentence to "ten forty five A M on twelfth of November twenty twenty six"; "AM" is spelled as two letters and "12" becomes the ordinal "twelfth". The word andaroo (Telugu for "everyone") is not in the dictionary, so the G2P model produces AE0 N D AA1 R OW0 ("an-DAA-roh"), a plausible reading of the spelling that was not checked against the actual pronunciation. The comma is rendered as a pause.

Telugu

The Telugu test text in the assignment PDF contains broken spacing and detached diacritics, apparently from copy-paste. The spacing was repaired for T1, T2a, T2b and T4. For T3, both the text as printed (అడ విలో) and the repaired text (అడవిలో) were synthesised (T3 and T3_joined).

T1 – న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

The first word is a long English loanword (neuropharmacological) written in Telugu script and mapped to 19 phones, including ph, which is rare in the training data (155 occurrences; au in ఔషధాల has 164). The mapping is consistent with the spelling. This sentence was expected to be the weakest because these sounds are rare.

T2a / T2b – a) తాడుపై అతని పట్టు చాలా బలంగా ఉంది. b) ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

పట్టు means "grip" in (a) and "silk" in (b). It is written and pronounced identically (p a tt tt u), so, unlike the English homograph test, there is no front-end decision and any difference can appear only in the prosody. Nasals are handled correctly: the anusvara assimilates to the following consonant (బలంగా → b a l a ng g aa, ఉంది → u n d i, ధరించి → dh a r i ny ch i).

T3 – పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

With the printed text, అడవిలో is split into అడ and విలో (a dd a / v i l oo). The phones are the same as in the joined version; only the word boundary differs, and it has no acoustic consequence (section 5). The long consonant శ in నిశ్శబ్దంగా is mapped to a doubled sh (sh sh).

T4 – ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Numerals are expanded into words before synthesis: 2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు. This is correct for the date and the time. The comma becomes a pause. With 120 phones this is the longest test input in either language.

5. Variance Adaptor: Duration, Pitch and Energy

Predicted duration, F0 and energy per phone were plotted for every test sentence (Results/<Language>/variance_adaptor_plots/, with the underlying numbers in the data/ folder). One frame is 256 samples at 22.05 kHz, about 11.6 ms. No sentence has a zero-length phone, and the only pauses are at commas. The figures show the longest sentence of each language (E4 and T4).

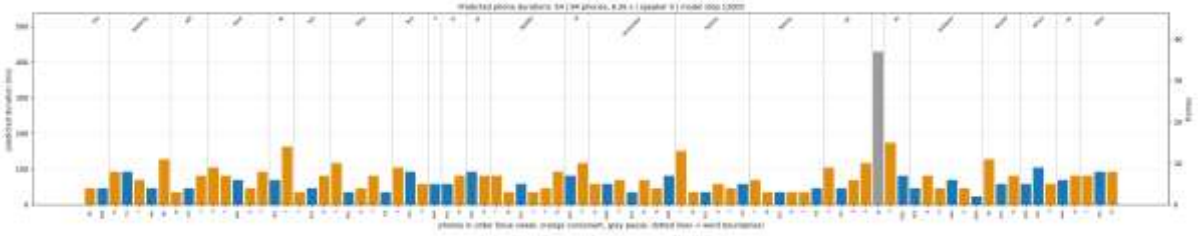


Figure 5a. E4, predicted duration per phone (the shaded block is the pause at the comma).

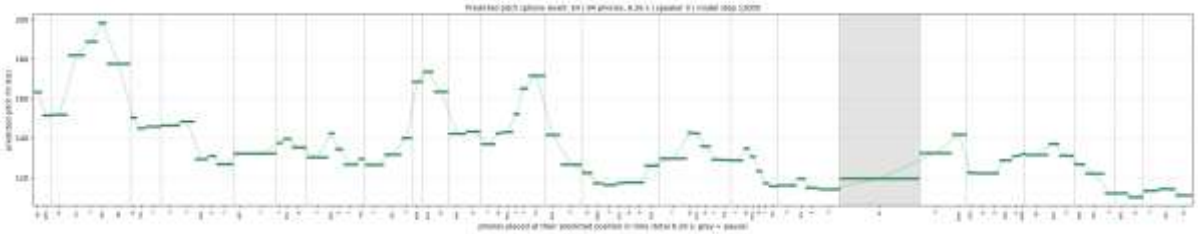


Figure 5b. E4, predicted F0.

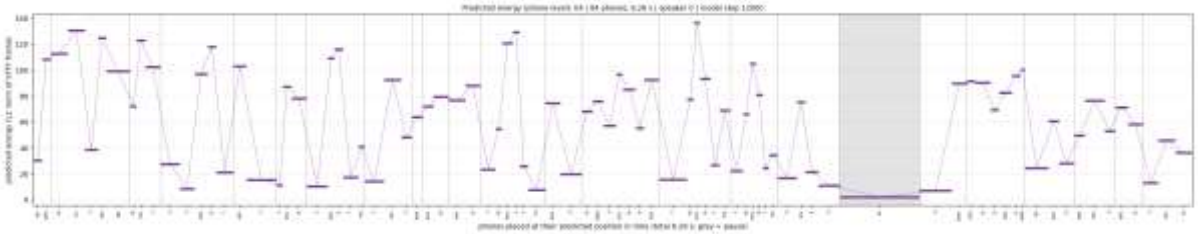


Figure 5c. E4, predicted energy.

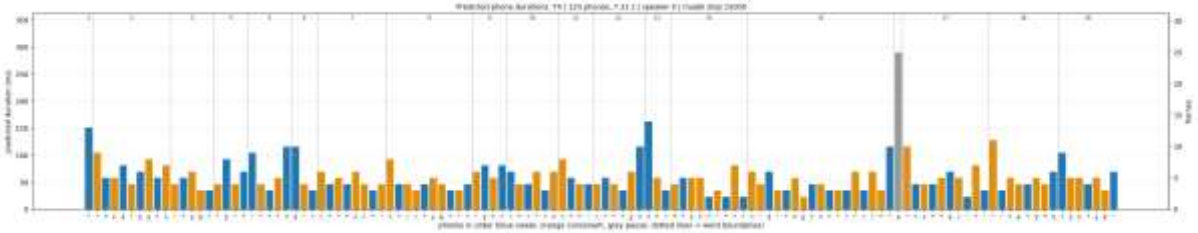


Figure 6a. T4, predicted duration per phone (the shaded block is the pause at the comma).

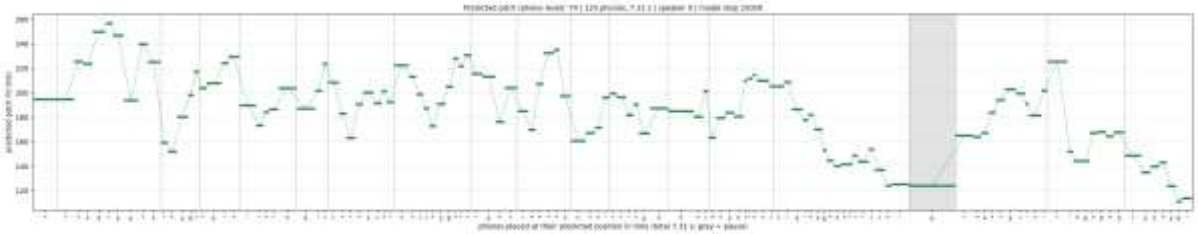


Figure 6b. T4, predicted F0.

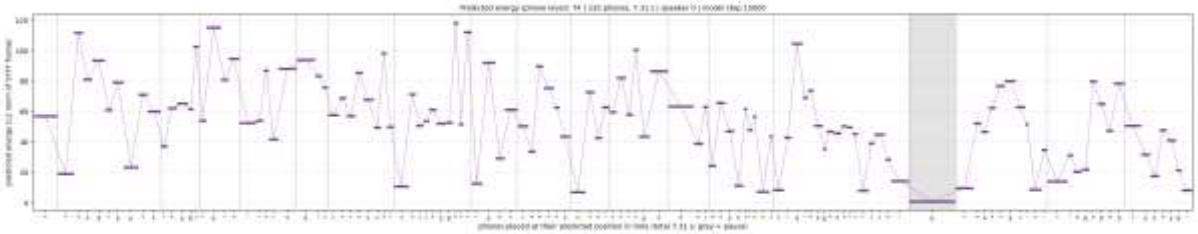


Figure 6c. T4, predicted energy.

ID	Phones	Length (s)	Mean phone (ms)	Pause (ms)	F0 first → last (Hz)	F0 min–max (Hz)	Energy peak on
E1	87	5.36	61.7	–	135 → 113	105–169	considerable
E2a	51	3.79	74.2	–	147 → 97	97–153	orchestra's
E2b	56	3.66	65.3	–	173 → 111	111–173	proposed
E3	37	2.57	69.3	–	140 → 115	115–156	the
E4	84	6.26	70.2	430	164 → 111	110–198	twenty
T1	66	4.64	70.4	–	134 → 111	109–280	న్యూరోఫార్మకోలాజికల్
T2a	31	2.14	68.9	–	169 → 128	126–228	చాలా
T2b	37	2.23	60.2	–	155 → 106	106–253	చీరను
T3	31	1.77	56.9	–	194 → 120	120–272	విలో
T4	120	7.31	59.0	290	195 → 114	111–257	పన్నెండవ

English

Common pattern. In all five sentences F0 starts high and ends at or near its lowest value (147→97 Hz in E2a, 173→111 Hz in E2b), the falling contour of a statement. Phrase-final lengthening is visible: the final syllable of “unexpectedly” in E3 (IY0, 255 ms) is the longest phone of that sentence. Mean phone duration is 62–74 ms. Energy ranges from about 6–11 at its lowest to 119–147 at its peak.

E1. No pause is predicted. “antidisestablishmentarianism” lasts 1.72 s (about a third of the sentence) at 61 ms per phone, equal to the sentence average, so the duration predictor scales with phone count and does not compress the long word. F0 rises from 120 Hz to the sentence peak of 169 Hz inside this word. The lowest F0 (105 Hz) and the energy peak (119) both fall in “considerable”.

E2a and E2b. The word “conduct” receives different prosody. As a noun in (a) it lasts 499 ms and F0 inside it spans 114–153 Hz, the F0 peak of the sentence. As a verb in (b) it lasts 453 ms with F0 between 133 and 149 Hz, below the sentence start. In (b) the F0 peak (173 Hz) is on the first word “the”, so the pitch predictor mainly applies a high start with a gradual decline rather than accents on content words. Sentence (a) is the slowest (74 ms per phone), with long fricatives such as the /s/ of “audience” (186 ms), and F0 falls by about 50 Hz.

E3 and E3_fixed. The split "rhyth m" lasts 197 + 186 = 383 ms against 337 ms for "rhythm", and the isolated "m" receives its own F0 peak (156 Hz), as if it were a stressed word. In E3_fixed the F0 peak moves to "of" (169 Hz) and the energy peak (147) to "rhythm", whereas in E3 the energy peak is on "the" (129). The broken input thus makes the predictors treat a stray letter as a word.

E4. This is the longest English sentence (6.26 s) and the only one with a pause (430 ms at the comma). F0 starts at 164 Hz and peaks at 198 Hz on "meeting", the highest of the English test sentences, and rises again on "A M" (164–174 Hz) and "twelfth" (up to 172 Hz), so the time and date are read with raised pitch. The energy peak (136) is on "twenty". After the comma F0 resets to 133–142 Hz on "so" and falls to 111 Hz, giving two intonation phrases.

Telugu

Common pattern. The contours have the same shape as in English (195→114 Hz in T4, 194→120 Hz in T3), but pitch is higher and more variable: mean F0 per sentence is 183–193 Hz with a standard deviation of 26–37 Hz, against 131–139 Hz and 11–19 Hz for English. Mean phone duration is 63 ms against 68 ms for English (averages over the five sentences; the phone inventories are not directly comparable). The dataset card lists the gender label with two classes, 0 = female, so label 1 is the male voice that was selected. A mean F0 near 185 Hz is high for a male speaker, which is consistent with possible octave errors in the PyWorld pitch tracks (section 6).

T1. The first word lasts 1.51 s (19 phones, about 79 ms per phone). F0 climbs inside it from 134 to 279 Hz, the highest F0 prediction in the test set, and the energy peak (131) is also in this word. This is probably the pitch predictor reacting to a word made of rare phones, in line with the pitch validation loss that did not improve. The following words descend from 182–242 Hz (పరిశోధనలో) to 109–181 Hz in the last word.

T2a and T2b. The word పట్టు has the same phones in both sentences and is realised almost identically: 302 ms with F0 185–228 Hz in (a) and 290 ms with 178–227 Hz in (b), as expected from a text-only front end. The differences lie in the sentence contour. In (a) the F0 peak (228 Hz) is on పట్టు and the energy peak is on the intensifier చాలా ("very"). In (b) the F0 peak is late, on వేడుకకు (253 Hz), and the energy peak (104, on చీరను) is the lowest of the Telugu sentences.

T3 and T3_joined. The summary statistics are identical for both versions (31 phones, 1.765 s, F0 range 120–272 Hz, no pause), so the extra space in అడ విలో has no effect on the output: it creates a word boundary in the text but no pause. This is the shortest and fastest sentence (56.9 ms per phone). F0 peaks at 272 Hz on the first word పులి, and the longest word is నిశ్శబ్దంగా (11 phones, 626 ms).

T4. One 290 ms pause occurs at the comma. F0 starts at 195 Hz, peaks at 257 Hz on సమావేశం, falls to about 125 Hz at the end of the first clause, resets to 164 Hz or higher after the comma and ends at 111–149 Hz on ఉంచండి, giving two intonation phrases as in English. The longest word, ప్రారంభమవుతుంది (17 phones), takes 859 ms (about 50 ms per phone, below the sentence average of 59 ms), so the duration predictor does compress a long word here. The numerals (రెండు వేల ఇరవై ఆరు) have F0 between 152 and 230 Hz.

6. English Compared with Telugu

The differences between the two models follow mainly from the data. Telugu is a single studio voice with a rule-based phone set and an aligner trained on the same audio; English is accented speech from many speakers aligned with a pretrained US-English aligner. The validation duration loss is 0.061 for

Telugu and 0.24 for English, and Telugu validation loss plateaus whereas English rises after step 12,000. Total losses are not directly comparable because the data and the normalised pitch and energy differ.

The predicted prosody also differs. English F0 contours are narrow (standard deviation 11–19 Hz around about 135 Hz); Telugu contours are wide (26–37 Hz around about 185 Hz) and include extreme single-word jumps such as 134→279 Hz in T1. This may reflect a more expressive speaker, octave errors in the PyWorld training pitch, or predictor overfitting; these causes were not separated.

Limitations and failures

- **Broken input.** In E3, "rhyth m" is read as two words and the audio is wrong. The same could happen to Telugu text with broken spacing, although the broken input in T3 did not change the output.
- **English speaker ids.** The 115 speakers are demographic profiles, not real people, so the speaker embedding has limited meaning; speaker 0 is the profile with the most data.
- **Overfitting.** English overfits after about 12,000 steps, and the pitch and energy predictors memorise the training set in both languages.
- **Rare and unseen words.** Out-of-vocabulary words (andaroo) depend on a G2P guess, and the rule-based Telugu front end makes loanwords with rare phones (T1) the weakest case.
- **Text normalisation.** Numbers and dates are handled by hand-written rules tested only on the assignment sentences.
- **Evaluation.** The data is small for FastSpeech 2 (4.3–4.7 hours) and no listening test or objective measure such as MCD was run, so quality statements beyond the plots are subjective.

Possible improvements

- Early stopping and stronger regularisation (more dropout, weight decay or a smaller model) for English.
- A text normaliser with unit tests that also removes stray spaces inside words.
- Real speaker ids for English, using a few large speakers or audio clustering.
- Smoothing F0 and removing octave jumps from the training pitch to stabilise the Telugu pitch predictor.
- A shared phone inventory so that all 9 hours can be used, and a listening test with MCD and ASR word error rate for objective quality.

7. Reproducibility

Software. Python 3, PyTorch, PyWorld, tgt, g2p_en, num2words, datasets and huggingface_hub (see requirements.txt), and Montreal Forced Aligner 3.x installed through conda-forge.

Structure. Each language folder contains preprocess.py, train.py, inference.py and config.yaml (paths, run name, number of steps, inference options). These scripts wrap the FastSpeech 2 core code in <Language>/fastspeech2/, which has its own model config.yaml; at run time the Kaggle paths in that file are rewritten to local folders.

Commands (from the repository root):

```
# once: put the core code in English/fastspeech2 and Telugu/fastspeech2
python tools/merge_code_zips.py English_code.zip Telugu_code.zip
export HF_TOKEN=hf_xxx

cd English        # then the same in Telugu/
python preprocess.py   # data selection (seed 42) -> alignment -> features
```

```
                        # (--from-hf reuses the prepared data and alignments)
python train.py         # 30,000 steps (Telugu 15,000), resumes automatically
python inference.py     # audio and plots -> ../Results/English
```

Checkpoints. English/checkpoints/best.pth.tar (step 12,000) and Telugu/checkpoints/best.pth.tar (step 15,000), about 142 MB each, are stored with Git LFS and are also publicly available on Hugging Face (English_run/best.pth.tar and Telugu_run/best.pth.tar): https://huggingface.co/datasets/akshaya148/tts-prepared-data/tree/main

Seed. 42 for data selection and the train/validation split.